



Enhancing Large Language Models for Combinatorial Discovery

Group 99
112550127 Pin-Kuan Chiang
Advisor: Ping-Chun Hsieh

Abstract

LLM-guided evolutionary search is promising for combinatorial discovery, but often suffers from diversity collapse and unstable evaluation. We propose **Rainbow EoH**, a diversity-preserving and robustly evaluated extension of EoH for Salem-Spencer set discovery.

Research Objectives

- Search Diversity:** Avoid collapse into structurally similar programs.
- Robust Evaluation:** Reduce single-scale and stochastic overfitting.

Main Contributions

- AST + islands for diversity preservation.
- Pass@K + cross- N for robust evaluation.
- Improve total score from **201** to **217**.

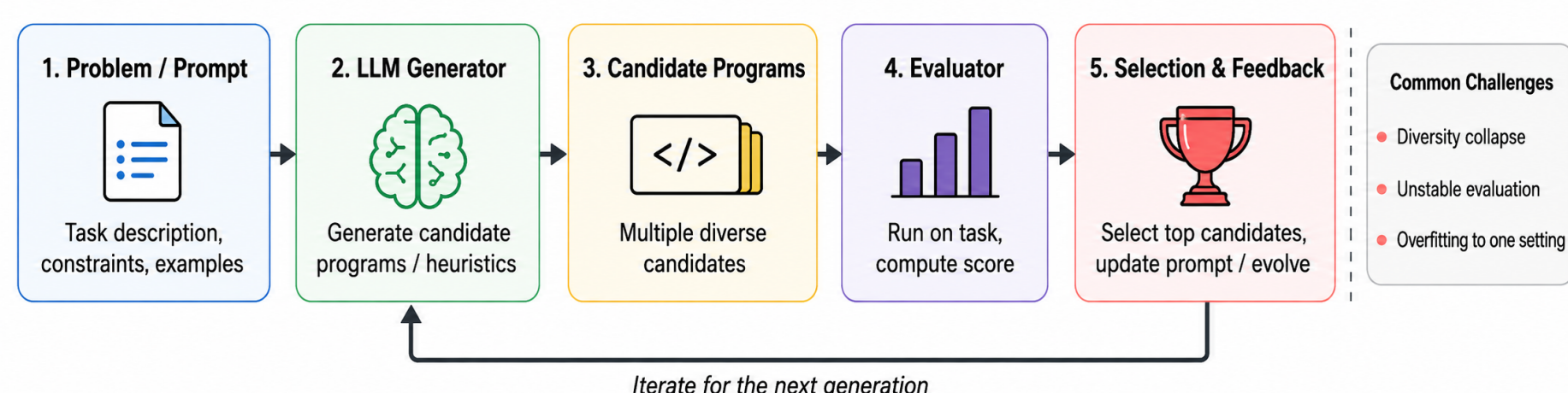
Background & Related Works

Salem-Spencer Sets

A Salem-Spencer set is a subset $A \subseteq \{1, 2, \dots, N\}$ containing no distinct $x, y, z \in A$ such that $x + z = 2y$ [1]. Behrend [2] gave the best-known lower bound; Roth [3] proved the first non-trivial upper bound, recently sharpened to near-optimal [4, 5]. For a computational survey see [6].

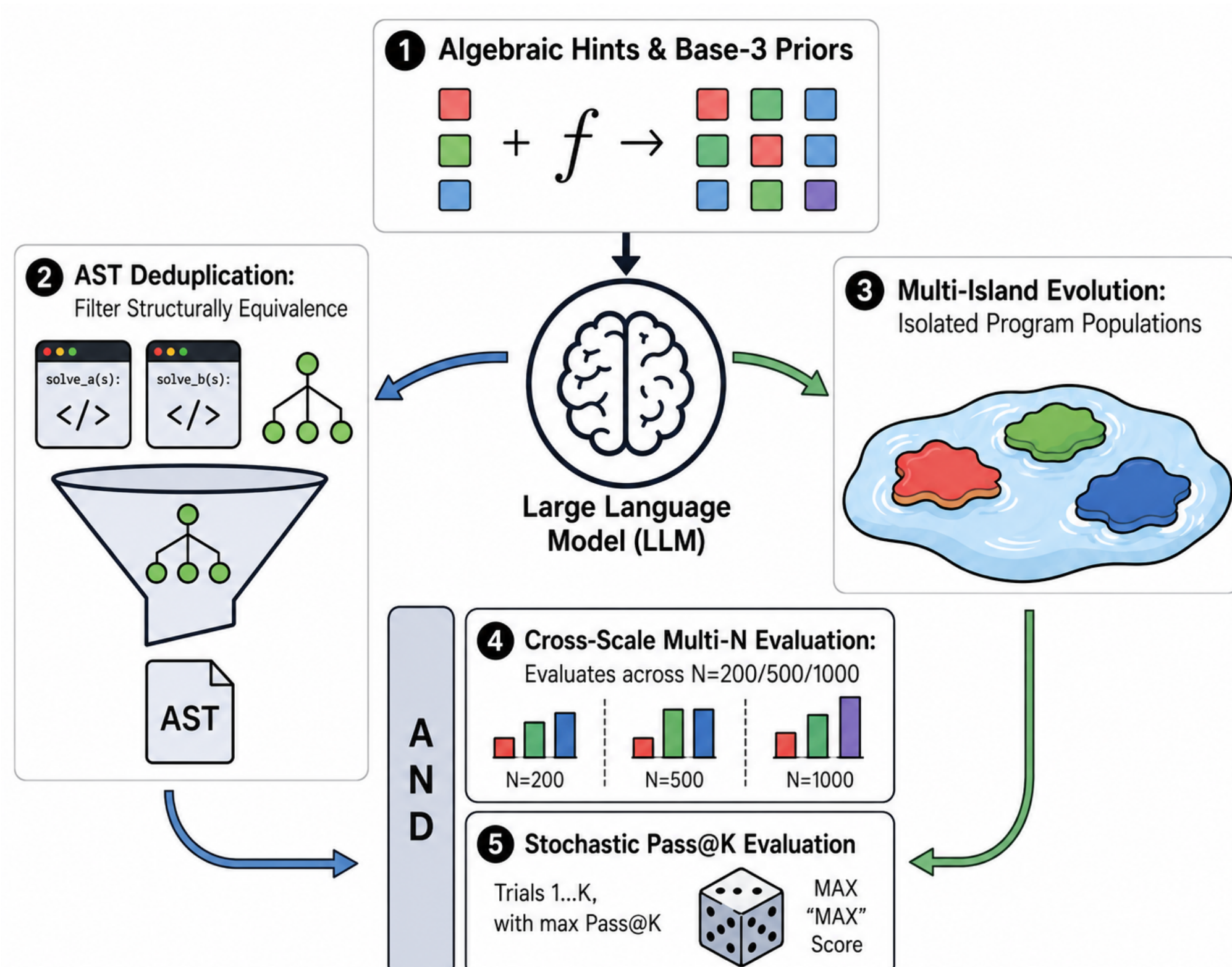
LLM-Driven Algorithm Discovery

LLMs propose candidate heuristics, which are evaluated and refined through evolutionary feedback [7, 8, 9]. Rainbow EoH builds directly on the EoH framework [8].



Methodology

Rainbow EoH improves LLM-guided evolutionary search by separating the pipeline into **generation guidance**, **diversity preservation**, and **robust evaluation**.



Preliminary Results & Analysis

Ablation Setup

We evaluate Rainbow EoH over three Salem-Spencer problem sizes: $N = 200, 500, 1000$. The objective is reported as $N=200/N=500/N=1000$, and the total score is their sum. All rows use the same multi- N protocol; the ablation focuses on four modules: algebraic prior, AST deduplication, island evolution, and stochastic Pass@K.

Setting	Prior	AST	Islands	Pass@K	Gen.	Best Obj.	Total
Baseline EoH	–	–	–	–	19	32/64/105	201
Prior only	on	–	–	–	21	32/64/105	201
AST only	–	on	–	–	19	32/64/105	201
Pass@K only	–	–	–	on	31	32/64/105	201
Islands only	–	–	on	–	19	32/64/105	201
Full Rainbow EoH	on	on	on	on	52	37/74/106	217
w/o Prior	–	on	on	on	43	37/74/106	217
w/o AST	on	–	on	on	15	32/64/105	201
w/o Pass@K	on	on	on	–	15	36/64/105	205
w/o Islands	on	on	–	on	18	36/64/105	205

Table 1. Ablation results. Prior = algebraic hints; AST = structural deduplication; Islands = isolated populations; Pass@K = stochastic multi-trial evaluation.

Key Findings

Full Rainbow EoH achieves the strongest result. The complete system improves the total score from 201 to 217, with the largest gains at $N=200$ and $N=500$.

Diversity mechanisms are effective only in combination. Individual modules do not improve the final objective, but their combination extends the search to 52 generations, compared with 19 in the baseline.

AST deduplication is the most critical module. Removing AST deduplication drops the score back to 201, while removing Pass@K or islands reduces the score to 205.

Implications and Future Work

These results suggest that Rainbow EoH improves combinatorial discovery mainly by preserving structural diversity and enabling longer exploration. Future work will study stronger algebraic priors, larger N , island migration, and repeated runs for variance estimation.

References

- [1] R. Salem and D. C. Spencer, "On sets of integers which contain no three terms in arithmetical progression," *Proceedings of the National Academy of Sciences of the United States of America*, vol. 28, no. 12, pp. 561–563, 1942.
- [2] F. A. Behrend, "On sets of integers which contain no three terms in arithmetical progression," *Proceedings of the National Academy of Sciences of the United States of America*, vol. 32, no. 12, pp. 331–332, 1946.
- [3] K. F. Roth, "On certain sets of integers," *Journal of the London Mathematical Society*, vol. s1-28, no. 1, pp. 104–109, 1953.
- [4] Z. Kelley and R. Meka, "Strong bounds for 3-progressions," *arXiv preprint arXiv:2302.05537*, 2023.
- [5] R. Raghavan, "Improved bounds for 3-progressions," *arXiv preprint arXiv:2603.27045*, 2026.
- [6] W. Gasarch, J. Glenn, and C. Kruskal, "Finding large sets without arithmetic progressions of length three: An empirical view and survey ii," *arXiv preprint arXiv:2501.01634*, 2025.
- [7] B. Romera-Paredes, M. Barekatin, A. Novikov, M. Balog, M. P. Kumar, E. Dupont, F. J. R. Ruiz, J. S. Ellenberg, P. Wang, O. Fawzi, P. Kohli, and A. Fawzi, "Mathematical discoveries from program search with large language models," *Nature*, vol. 625, no. 7995, pp. 468–475, 2024.
- [8] F. Liu, X. Tong, M. Yuan, X. Lin, F. Luo, Z. Wang, Z. Lu, and Q. Zhang, "Evolution of heuristics: Towards efficient automatic algorithm design using large language model," in *Proceedings of the 41st International Conference on Machine Learning*, vol. 235 of *Proceedings of Machine Learning Research*, pp. 32201–32223, PMLR, 2024.
- [9] H. Ye, J. Wang, Z. Cao, F. Berto, C. Hua, H. Kim, J. Park, and G. Song, "Reevo: Large language models as hyper-heuristics with reflective evolution," *arXiv preprint arXiv:2402.01145*, 2024.